

DERIVED METRICS, PHYSIOLOGIC INTERPRETATION, AND LONGITUDINAL ANALYTICS

Derived Physiology Concept

The wearable longitudinal penile physiology monitoring device may generate derived metrics based on interpretation of raw sensor outputs rather than relying solely on direct measurements. Mechanical, thermal, optical, inertial, and contextual sensing channels may be computationally combined to infer physiologic states, transitions, and behavioral patterns across time. Derived metrics may represent composite physiologic indicators reflecting multidimensional analysis rather than isolated sensor readings.

Erection Onset and Rise Dynamics

Derived analytics may include identification of erection onset timing, rate of rigidity increase, time-to-threshold measurements, slope of rigidity development, acceleration characteristics, and stabilization behavior. These parameters may be calculated using deformation sensing, perfusion indicators, temperature trends, and motion-aware validation to characterize dynamic physiologic response patterns.

Rigidity Stability and Duration Metrics

Longitudinal analysis may quantify duration above defined rigidity thresholds, stability of maintained erection states, variability during sustained episodes, and temporal continuity of physiologic activation. Metrics may include area-under-curve representations, consistency indices, micro-variability analysis, and episode fragmentation detection.

Detumescence and Recovery Characterization

Detumescence behavior may be analyzed through rate of rigidity decline, recovery timing, transition smoothness, and return-to-baseline characteristics. Computational interpretation may classify detumescence patterns including gradual decay, rapid collapse, interrupted episodes, or multi-phase recovery responses.

Nocturnal Physiology and Sleep-Associated Events

During rest or sleep periods, the device may identify nocturnal erection events, episode frequency, timing distribution, peak rigidity levels, and cumulative physiologic activation duration. Integration with inertial sensing may allow low-motion validation to improve confidence in sleep-associated physiologic interpretations.

Contextual Confidence and Quality Indices

Sensor fusion algorithms may generate confidence scores representing reliability of detected physiologic events. Contextual indices may incorporate motion stability, temperature conditions, signal quality indicators, and sensor agreement metrics to produce quality assessments associated with individual episodes or longitudinal trends.

Longitudinal Trend Analysis

Aggregated measurements collected across extended periods may enable trend analysis including baseline establishment, variability monitoring, recovery tracking, adaptation patterns, and comparative temporal analysis. Longitudinal analytics may be performed locally or within associated computational systems to generate user-specific physiologic profiles.

The analytic embodiments described herein are illustrative and non-limiting and encompass present and future computational techniques capable of deriving physiologic insight from multi-modal longitudinal sensing data.